# symGCAD / corpus benchmark

303 / 331 distinct original problems independently verified · 28 marked slow

All distinct problems are listed below, fastest verified solve first; unresolved problems follow. **Solve** and **check** are fresh-process wall seconds. **RAM** is maximum recorded resident MiB across solve and verification (†: verification RSS unavailable). **N** is the repetition count; three-run medians are preferred when available. **D**/**R** identifies debug/release builds. Cache tags: **C** = cold; **S** = shared checkpoints (may be warm); **?** = unrecorded. For incomplete rows, **TO**/**stop** is the longest retained observed attempt, not a solved time. Their RAM is the highest retained attempt RSS, which may come from an earlier run. The **baseline** columns are a separate, matched GCAD comparison: projection + lifting + merging only, with three fresh repetitions and identical orders. ± means two separate sign-branch solves, not the optimized full-split solve.

Matched ratio = symGCAD / GCAD: < 1 1–2 > 2 — = unavailable or censored; TO = native timeout. Ratios use two significant digits.

| # | Original problem / representative full ID | IDs | Method · build | N | Solve s | Check s | RAM MiB | Cells | Matched Rust s (×R) | GCAD s |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | test_input_unmodified/solve/4 | 1 | optimized<br>9025e71 R · ? | 1 | 0.0412 | 0.0211 | 6.06 | 2 | 0.000076 **(×0.045)** | 0.0017 |
| 2 | finiteA_3_25_4_0__2_0_0_1_1_0_0_0_0/symbolic | 1 | monotone+H<br>9025e71 R · ? | 1 | 0.0413 | 0.0210 | 6.05 | 1 | 0.0011 **(×0.095)** ± | 0.011 |
| 3 | finiteA_3_21_4_0__2_0_1_0_1_0_0_0_0/symbolic | 1 | monotone+H<br>9025e71 R · ? | 1 | 0.0414 | 0.0216 | 6.07 | 5 | 0.00063 **(×0.053)** ± | 0.012 |
| 4 | test_ex44_B1_C2/solve/1 | 1 | optimized<br>9025e71 R · ? | 1 | 0.0427 | 0.0210 | 6.06 | 0 | 0.0077 **(×0.47)** | 0.016 |
| 5 | test_cylinder_3d/solve/1 | 1 | optimized<br>9025e71 R · ? | 1 | 0.0613 | 0.0211 | 6.02 | 1 | 0.00037 **(×0.099)** | 0.0037 |
| 6 | test_greedy_sotd_order/solve/2 | 1 | optimized<br>9025e71 R · ? | 1 | 0.0613 | 0.0210 | 6.04 | 7 | 0.00049 **(×0.11)** | 0.0047 |
| 7 | test_greedy_sotd_order/order/5 | 1 | optimized<br>9025e71 R · ? | 1 | 0.0613 | 0.0209 | 6.03 | 9 | 0.00067 **(×0.14)** | 0.0049 |
| 8 | test_empty_3d/solve/1 | 1 | optimized<br>9025e71 R · ? | 1 | 0.0613 | 0.0210 | 6.07 | 0 | 0.00013 **(×0.062)** | 0.0020 |
| 9 | test_no_real_roots/solve/1 | 1 | optimized<br>9025e71 R · ? | 1 | 0.0613 | 0.0210 | 4.46 | 1 | 0.00020 **(×0.16)** | 0.0013 |
| 10 | test_unbounded_neg/solve/1 | 1 | optimized<br>9025e71 R · ? | 1 | 0.0613 | 0.0209 | 4.46 | 1 | 0.000048 **(×0.036)** | 0.0013 |
| 11 | finiteE_1_1__3_0__3_0_0_0/symbolic | 1 | monotone+H<br>9025e71 R · ? | 1 | 0.0613 | 0.0209 | 6.06 | 1 | 0.00017 **(×0.043)** ± | 0.0038 |
| 12 | test_greedy_sotd_order/solve/1 | 1 | optimized<br>9025e71 R · ? | 1 | 0.0613 | 0.0210 | 6.02 | 9 | 0.00045 **(×0.095)** | 0.0047 |
| 13 | test_greedy_sotd_order/order/3 | 1 | optimized<br>9025e71 R · ? | 1 | 0.0613 | 0.0210 | 6.04 | 9 | 0.00046 **(×0.094)** | 0.0049 |
| 14 | test_x2_minus_y/solve/1 | 1 | optimized<br>9025e71 R · ? | 1 | 0.0613 | 0.0210 | 4.52 | 1 | 0.000065 **(×0.044)** | 0.0015 |
| 15 | test_quadrant/solve/1 | 1 | optimized+H<br>9025e71 R · ? | 1 | 0.0614 | 0.0210 | 4.57 | 1 | 0.000078 **(×0.051)** | 0.0015 |
| 16 | test_y2_minus_x/solve/1 | 1 | optimized<br>9025e71 R · ? | 1 | 0.0614 | 0.0210 | 4.54 | 1 | 0.00015 **(×0.088)** | 0.0017 |
| 17 | test_greedy_mods_order/order/1 | 1 | optimized<br>9025e71 R · ? | 1 | 0.0614 | 0.0211 | 6.03 | 6 | 0.00014 **(×0.045)** | 0.0031 |
| 18 | test_ex422_bubble/solve/1 | 1 | optimized+H<br>9025e71 R · ? | 1 | 0.0614 | 0.0211 | 6.04 | 2 | 0.00032 **(×0.092)** | 0.0034 |
| 19 | test_greedy_sotd_order/order/4 | 1 | optimized<br>9025e71 R · ? | 1 | 0.0614 | 0.0210 | 6.04 | 7 | 0.00046 **(×0.098)** | 0.0047 |

| # | Original problem / representative full ID | IDs | Method · build | N | Solve s | Check s | RAM MiB | Cells | Matched Rust s (×R) | GCAD s |
|---|---|---|---|---|---|---|---|---|---|---|
| 20 | finiteA_4_85_5_0__2_0_1_0_1_0_1_0_0/symbolic | 1 | monotone+H<br>9025e71 R · ? | 1 | 0.0614 | 0.0211 | 6.07 | 15 | 0.0026 **(×0.075)** ± | 0.035 |
| 21 | finiteA_4_154_5_0__0_2_0_1_1_0_0_1_0/symbolic | 1 | monotone+H<br>9025e71 R · ? | 1 | 0.0614 | 0.0211 | 6.05 | 8 | 0.0046 **(×0.095)** ± | 0.048 |
| 22 | test_x_y_z_minus_1/solve/1 | 1 | optimized<br>9025e71 R · ? | 1 | 0.0614 | 0.0211 | 6.05 | 2 | 0.00020 **(×0.039)** | 0.0052 |
| 23 | test_hypercube_2/solve/1 | 1 | optimized<br>9025e71 R · ? | 1 | 0.0614 | 0.0215 | 6.03 | 1 | 0.00016 **(×0.038)** | 0.0043 |
| 24 | finiteAx12_4_27_5_0__2_1_0_1_1_0_0_0_0/symbolic | 1 | monotone+H<br>9025e71 R · ? | 1 | 0.0615 | 0.0212 | 6.07 | 9 | 0.0026 **(×0.085)** ± | 0.031 |
| 25 | finiteAx12_4_154_5_0__0_2_0_1_1_0_0_1_0/symbolic | 1 | monotone+H<br>9025e71 R · ? | 1 | 0.0615 | 0.0211 | 6.07 | 8 | 0.0037 **(×0.11)** ± | 0.034 |
| 26 | finiteAx12_4_312_5_0__0_0_0_2_1_1_0_0_1/symbolic | 1 | monotone+H<br>9025e71 R · ? | 1 | 0.0615 | 0.0211 | 6.06 | 9 | 0.0019 **(×0.075)** ± | 0.026 |
| 27 | finiteB_4_85_5_0__2_0_1_0_1_0_1_0_0/symbolic | 1 | monotone+H<br>9025e71 R · ? | 1 | 0.0615 | 0.0211 | 6.05 | 3 | 0.0015 **(×0.084)** ± | 0.018 |
| 28 | finiteAx12_3_26_4_0__0_2_0_1_1_0_0_0_0/symbolic | 1 | monotone+H<br>9025e71 R · ? | 1 | 0.0615 | 0.0211 | 6.07 | 8 | 0.00085 **(×0.063)** ± | 0.013 |
| 29 | test_input_unmodified/order/1 | 1 | optimized<br>9025e71 R · ? | 1 | 0.0615 | 0.0211 | 6.06 | 2 | 0.00013 **(×0.075)** | 0.0017 |
| 30 | test_hypercube_5/solve/1 | 1 | optimized<br>9025e71 R · ? | 1 | 0.0615 | 0.0211 | 6.04 | 2 | 0.00058 **(×0.098)** | 0.0060 |
| 31 | finiteA_4_293_5_0__2_0_1_0_0_1_0_0_1/symbolic | 3 | monotone+H<br>9025e71 R · ? | 1 | 0.0615 | 0.0212 | 6.07 | 6 | 0.0014 **(×0.068)** ± | 0.021 |
| 32 | finiteA_4_89_5_0__2_0_0_1_1_0_1_0_0/symbolic | 1 | monotone+H<br>9025e71 R · ? | 1 | 0.0615 | 0.0211 | 6.05 | 8 | 0.0022 **(×0.069)** ± | 0.032 |
| 33 | finiteA_4_23_5_0__2_1_1_0_1_0_0_0_0/symbolic | 1 | monotone+H<br>9025e71 R · ? | 1 | 0.0615 | 0.0212 | 6.07 | 6 | 0.0010 **(×0.056)** ± | 0.018 |
| 34 | finiteE_2_10__3_0__0_2_0_1/symbolic | 2 | monotone+H<br>9025e71 R · ? | 1 | 0.0615 | 0.0211 | 6.07 | 5 | 0.0012 **(×0.10)** ± | 0.011 |
| 35 | test_input_unmodified/order/2 | 1 | optimized<br>9025e71 R · ? | 1 | 0.0615 | 0.0211 | 6.04 | 2 | 0.000097 **(×0.047)** | 0.0021 |
| 36 | finiteA_4_297_5_0__2_0_0_1_0_1_0_0_1/symbolic | 3 | monotone+H<br>9025e71 R · ? | 1 | 0.0615 | 0.0212 | 6.07 | 9 | 0.0039 **(×0.088)** ± | 0.044 |
| 37 | finiteAx12_4_297_5_0__2_0_0_1_0_1_0_0_1/symbolic | 3 | monotone+H<br>9025e71 R · ? | 1 | 0.0615 | 0.0237 | 6.06 | 9 | 0.0033 **(×0.097)** ± | 0.034 |
| 38 | finiteA_4_278_5_0__0_2_1_0_1_0_0_0_1/symbolic | 1 | monotone+H<br>9025e71 R · ? | 1 | 0.0615 | 0.0211 | 6.07 | 5 | 0.0010 **(×0.053)** ± | 0.019 |
| 39 | finiteB_4_197_5_0__2_0_1_0_0_0_1_1_0/symbolic | 1 | monotone+H<br>9025e71 R · ? | 1 | 0.0615 | 0.0211 | 6.07 | 13 | 0.0012 **(×0.062)** ± | 0.019 |
| 40 | finiteA_5_309_6_0__2_0_1_0_1_1_0_0_1/symbolic | 2 | monotone+H<br>9025e71 R · ? | 1 | 0.0615 | 0.0212 | 6.06 | 42 | 0.40 **(×0.78)** ± | 0.51 |
| 41 | test_triangle2l_split/solve/3 | 1 | optimized+H<br>9025e71 R · ? | 1 | 0.0615 | 0.0211 | 6.05 | 6 | 0.0014 **(×0.086)** | 0.016 |
| 42 | finiteE_3_7__3_0__1_1_1_0/symbolic | 1 | monotone+H<br>9025e71 R · ? | 1 | 0.0615 | 0.0211 | 6.07 | 4 | 0.00066 **(×0.059)** ± | 0.011 |
| 43 | finiteA_4_27_5_0__2_1_0_1_1_0_0_0_0/symbolic | 1 | monotone+H<br>9025e71 R · ? | 1 | 0.0615 | 0.0211 | 6.06 | 9 | 0.0029 **(×0.082)** ± | 0.036 |
| 44 | test_triangle2l_split/order/1 | 2 | optimized+H<br>9025e71 R · ? | 1 | 0.0615 | 0.0211 | 6.05 | 4 | 0.0014 **(×0.081)** | 0.017 |
| 45 | test_ex411_box/solve/2 | 1 | optimized+H<br>9025e71 R · ? | 1 | 0.0615 | 0.0211 | 6.06 | 1 | 0.00018 **(×0.037)** | 0.0048 |

| # | Original problem / representative full ID | IDs | Method · build | N | Solve s | Check s | RAM MiB | Cells | Matched Rust s (×R) | GCAD s |
|---|---|---|---|---|---|---|---|---|---|---|
| 46 | finiteAx12_4_282_5_0__0_2_0_1_1_0_0_0_1/symbolic | 3 | monotone+H<br>9025e71 R · ? | 1 | 0.0615 | 0.0211 | 6.06 | 36 | 0.0032 **(×0.094)** ± | 0.034 |
| 47 | test_ex44_B1_B4/solve/1 | 1 | optimized<br>9025e71 R · ? | 1 | 0.0615 | 0.0211 | 6.05 | 0 | 0.00091 **(×0.12)** | 0.0074 |
| 48 | test_input_unmodified/order/3 | 1 | optimized<br>9025e71 R · ? | 1 | 0.0615 | 0.0212 | 6.07 | 2 | 0.000099 **(×0.047)** | 0.0021 |
| 49 | finiteA_4_284_5_0__0_0_2_1_1_0_0_0_1/symbolic | 3 | monotone+H<br>9025e71 R · ? | 1 | 0.0615 | 0.0212 | 6.05 | 5 | 0.0026 **(×0.080)** ± | 0.032 |
| 50 | finiteAx12_3_296_4_0__0_0_0_2_0_1_0_0_1/symbolic | 2 | monotone+H<br>9025e71 R · ? | 1 | 0.0615 | 0.0211 | 6.07 | 6 | 0.00082 **(×0.061)** ± | 0.014 |
| 51 | finiteA_3_26_4_0__0_2_0_1_1_0_0_0_0/symbolic | 1 | monotone+H<br>9025e71 R · ? | 1 | 0.0615 | 0.0211 | 6.06 | 8 | 0.0018 **(×0.11)** ± | 0.017 |
| 52 | test_ex411_box/solve/1 | 1 | optimized<br>9025e71 R · ? | 1 | 0.0615 | 0.0211 | 6.09 | 1 | 0.00016 **(×0.034)** | 0.0048 |
| 53 | test_hypercube_3/solve/1 | 1 | optimized<br>9025e71 R · ? | 1 | 0.0615 | 0.0367 | 6.05 | 1 | 0.00019 **(×0.038)** | 0.0050 |
| 54 | finiteB_5_341_6_0__2_0_1_0_1_0_1_0_1/symbolic | 3 | monotone+H<br>9025e71 R · ? | 1 | 0.0615 | 0.0211 | 6.05 | 11 | 0.0065 **(×0.085)** ± | 0.077 |
| 55 | finiteB_5_213_6_0__2_0_1_0_1_0_1_1_0/symbolic | 1 | monotone+H<br>9025e71 R · ? | 1 | 0.0615 | 0.0212 | 6.05 | 10 | 0.0028 **(×0.073)** ± | 0.038 |
| 56 | finiteAx12_3_296_4_0__0_0_0_2_0_1_0_0_1/fixed | 2 | monotone+H<br>9025e71 D · ? | 1 | 0.0615 | 0.0211 | 12.1 | 1 | 0.00079 **(×0.096)** ± | 0.0082 |
| 57 | finiteA_5_87_6_0__2_1_1_0_1_0_1_0_0/symbolic | 1 | monotone+H<br>9025e71 R · ? | 1 | 0.0615 | 0.0212 | 6.06 | 28 | 0.0056 **(×0.088)** ± | 0.064 |
| 58 | finiteA_4_344_5_0__0_0_0_2_1_0_1_0_1/symbolic | 3 | monotone+H<br>9025e71 R · ? | 1 | 0.0616 | 0.0212 | 6.05 | 10 | 0.0022 **(×0.071)** ± | 0.030 |
| 59 | finiteB_5_279_6_0__2_1_1_0_1_0_0_0_1/symbolic | 2 | monotone+H<br>9025e71 R · ? | 1 | 0.0616 | 0.0213 | 6.43 | 18 | 0.0061 **(×0.11)** ± | 0.054 |
| 60 | finiteA_4_23_5_0__2_1_1_0_1_0_0_0_0/fixed | 1 | monotone+H<br>9025e71 D · ? | 1 | 0.0616 | 0.0211 | 12.1 | 6 | 0.00083 **(×0.059)** ± | 0.014 |
| 61 | finiteA_5_121_6_0__2_0_0_1_1_1_1_0_0/symbolic | 1 | monotone+H<br>9025e71 R · ? | 1 | 0.0616 | 0.0212 | 6.04 | 9 | 0.0038 **(×0.071)** ± | 0.053 |
| 62 | finiteA_2_17_3_0__2_0_0_0_1_0_0_0_0/fixed | 1 | monotone+H<br>9025e71 D · ? | 1 | 0.0616 | 0.0211 | 12.1 | 1 | 0.00017 **(×0.040)** ± | 0.0043 |
| 63 | finiteAx12_4_27_5_0__2_1_0_1_1_0_0_0_0/fixed | 1 | monotone+H<br>9025e71 D · ? | 1 | 0.0616 | 0.0211 | 12.1 | 1 | 0.0011 **(×0.063)** ± | 0.018 |
| 64 | finiteE_3_11__3_0__1_1_0_1/symbolic | 2 | monotone+H<br>9025e71 R · ? | 1 | 0.0616 | 0.0211 | 6.07 | 8 | 0.0021 **(×0.093)** ± | 0.023 |
| 65 | finiteA_4_282_5_0__0_2_0_1_1_0_0_0_1/symbolic | 3 | monotone+H<br>9025e71 R · ? | 1 | 0.0616 | 0.0211 | 6.04 | 36 | 0.0037 **(×0.093)** ± | 0.040 |
| 66 | test_melih_box/solve/3 | 1 | optimized+H<br>9025e71 R · ? | 1 | 0.0616 | 0.0212 | 6.43 | 3 | 0.0028 **(×0.081)** | 0.034 |
| 67 | finiteA_4_153_5_0__2_0_0_1_1_0_0_1_0/fixed | 1 | monotone+H<br>9025e71 D · ? | 1 | 0.0616 | 0.0212 | 12.1 | 1 | 0.0021 **(×0.088)** ± | 0.024 |
| 68 | finiteE_4_15__4_0__1_1_1_1/fixed | 2 | monotone+H<br>9025e71 D · ? | 1 | 0.0616 | 0.0211 | 12.1 | 5 | 0.0021 **(×0.085)** ± | 0.025 |
| 69 | finiteA_5_310_6_0__0_2_1_0_1_1_0_0_1/symbolic | 1 | monotone+H<br>9025e71 R · ? | 1 | 0.0616 | 0.0211 | 6.05 | 6 | 0.0024 **(×0.064)** ± | 0.037 |
| 70 | finiteA_3_296_4_0__0_0_0_2_0_1_0_0_1/fixed | 2 | monotone+H<br>9025e71 D · ? | 1 | 0.0616 | 0.0212 | 12.1 | 1 | 0.00073 **(×0.077)** ± | 0.0095 |
| 71 | finiteB_3_21_4_0__2_0_1_0_1_0_0_0_0/fixed | 1 | monotone+H<br>9025e71 D · ? | 1 | 0.0616 | 0.0212 | 12.1 | 3 | 0.00030 **(×0.045)** ± | 0.0067 |

| # | Original problem / representative full ID | IDs | Method · build | N | Solve s | Check s | RAM MiB | Cells | Matched Rust s (×R) | GCAD s |
|---|---|---|---|---|---|---|---|---|---|---|
| 72 | finiteE_2_10__3_0__0_2_0_1/fixed | 2 | monotone+H<br>9025e71 D · ? | 1 | 0.0616 | 0.0212 | 12.1 | 1 | 0.00070 **(×0.12)** ± | 0.0059 |
| 73 | finiteE_3_7__3_0__1_1_1_0/fixed | 1 | monotone+H<br>9025e71 D · ? | 1 | 0.0616 | 0.0212 | 12.1 | 4 | 0.00063 **(×0.070)** ± | 0.0090 |
| 74 | finiteA_3_276_4_0__0_0_2_0_1_0_0_0_1/fixed | 2 | monotone+H<br>9025e71 D · ? | 1 | 0.0616 | 0.0212 | 12.1 | 4 | 0.00098 **(×0.12)** ± | 0.0083 |
| 75 | finiteA_4_312_5_0__0_0_0_2_1_1_0_0_1/symbolic | 1 | monotone+H<br>9025e71 R · ? | 1 | 0.0616 | 0.0211 | 6.06 | 9 | 0.0022 **(×0.076)** ± | 0.028 |
| 76 | finiteA_3_21_4_0__2_0_1_0_1_0_0_0_0/fixed | 1 | monotone+H<br>9025e71 D · ? | 1 | 0.0616 | 0.0212 | 12.1 | 5 | 0.00095 **(×0.12)** ± | 0.0080 |
| 77 | finiteC_5_158_6_0__0_2_1_1_1_0_0_1_0/fixed | 3 | monotone+H<br>9025e71 D · ? | 1 | 0.0616 | 0.0212 | 12.1 | 1 | 0.0047 **(×0.10)** ± | 0.046 |
| 78 | finiteC_5_214_10_0__0_6_1_0_1_0_1_1_0/fixed | 5 | monotone+H<br>9025e71 D · ? | 1 | 0.0616 | 0.0212 | 12.1 | 1 | 0.019 **(×0.19)** ± | 0.10 |
| 79 | finiteA_3_25_4_0__2_0_0_1_1_0_0_0_0/fixed | 1 | monotone+H<br>9025e71 D · ? | 1 | 0.0616 | 0.0212 | 12.1 | 1 | 0.00091 **(×0.096)** ± | 0.0095 |
| 80 | finiteA_4_278_5_0__0_2_1_0_1_0_0_0_1/fixed | 1 | monotone+H<br>9025e71 D · ? | 1 | 0.0616 | 0.0211 | 12.1 | 5 | 0.0010 **(×0.068)** ± | 0.015 |
| 81 | finiteF_4_15__4_0__1_1_1_1/fixed | 1 | monotone+H<br>9025e71 D · ? | 1 | 0.0616 | 0.0212 | 12.1 | 1 | 0.0022 **(×0.085)** ± | 0.026 |
| 82 | test_ex412_box/solve/1 | 1 | optimized<br>9025e71 R · ? | 1 | 0.0616 | 0.0212 | 6.50 | 2 | 0.00022 **(×0.034)** | 0.0065 |
| 83 | finiteA_4_27_5_0__2_1_0_1_1_0_0_0_0/fixed | 1 | monotone+H<br>9025e71 D · ? | 1 | 0.0616 | 0.0212 | 12.1 | 1 | 0.0012 **(×0.065)** ± | 0.019 |
| 84 | finiteAx12_4_154_5_0__0_2_0_1_1_0_0_1_0/fixed | 1 | monotone+H<br>9025e71 D · ? | 1 | 0.0616 | 0.0212 | 12.1 | 1 | 0.0022 **(×0.097)** ± | 0.023 |
| 85 | finiteA_4_297_5_0__2_0_0_1_0_1_0_0_1/fixed | 3 | monotone+H<br>9025e71 D · ? | 1 | 0.0616 | 0.0211 | 12.1 | 1 | 0.0017 **(×0.085)** ± | 0.021 |
| 86 | finiteE_2_9__3_0__2_0_0_1/fixed | 1 | monotone+H<br>9025e71 D · ? | 1 | 0.0616 | 0.0213 | 12.1 | 1 | 0.00082 **(×0.14)** ± | 0.0059 |
| 87 | finiteA_5_171_6_0__2_1_0_1_0_1_0_1_0/fixed | 1 | monotone+H<br>9025e71 D · ? | 1 | 0.0616 | 0.0212 | 12.1 | 1 | 0.0044 **(×0.082)** ± | 0.054 |
| 88 | finiteA_3_276_4_0__0_0_2_0_1_0_0_0_1/symbolic | 2 | monotone+H<br>9025e71 R · ? | 1 | 0.0616 | 0.0211 | 6.07 | 4 | 0.00075 **(×0.064)** ± | 0.012 |
| 89 | finiteC_5_227_6_0__2_1_0_0_0_1_1_1_0/fixed | 3 | monotone+H<br>9025e71 D · ? | 1 | 0.0616 | 0.0213 | 12.1 | 1 | 0.0063 **(×0.11)** ± | 0.056 |
| 90 | finiteA_5_410_10_0__0_6_0_1_1_0_0_1_1/fixed | 4 | monotone+H<br>9025e71 D · ? | 1 | 0.0616 | 0.0212 | 12.1 | 1 | 0.0085 **(×0.10)** ± | 0.084 |
| 91 | finiteAx12_4_297_5_0__2_0_0_1_0_1_0_0_1/fixed | 3 | monotone+H<br>9025e71 D · ? | 1 | 0.0617 | 0.0212 | 12.1 | 1 | 0.0017 **(×0.081)** ± | 0.020 |
| 92 | finiteAx12_3_26_4_0__0_2_0_1_1_0_0_0_0/fixed | 1 | monotone+H<br>9025e71 D · ? | 1 | 0.0617 | 0.0211 | 12.1 | 1 | 0.00090 **(×0.11)** ± | 0.0082 |
| 93 | finiteAx12_4_170_5_0__0_2_0_1_0_1_0_1_0/fixed | 1 | monotone+H<br>9025e71 D · ? | 1 | 0.0617 | 0.0212 | 12.1 | 1 | 0.0020 **(×0.096)** ± | 0.021 |
| 94 | finiteA_4_282_5_0__0_2_0_1_1_0_0_0_1/fixed | 3 | monotone+H<br>9025e71 D · ? | 1 | 0.0617 | 0.0211 | 12.1 | 1 | 0.0017 **(×0.082)** ± | 0.021 |
| 95 | finiteE_3_13__3_0__1_0_1_1/fixed | 1 | monotone+H<br>9025e71 D · ? | 1 | 0.0617 | 0.0212 | 12.1 | 4 | 0.0013 **(×0.10)** ± | 0.012 |
| 96 | finiteA_5_286_6_0__0_2_1_1_1_0_0_0_1/fixed | 3 | monotone+H<br>9025e71 D · ? | 1 | 0.0617 | 0.0212 | 27.5 | 6 | 0.0045 **(×0.091)** ± | 0.049 |
| 97 | finiteA_4_312_5_0__0_0_0_2_1_1_0_0_1/fixed | 1 | monotone+H<br>9025e71 D · ? | 1 | 0.0617 | 0.0212 | 12.1 | 1 | 0.0018 **(×0.096)** ± | 0.018 |

| # | Original problem / representative full ID | IDs | Method · build | N | Solve s | Check s | RAM MiB | Cells | Matched Rust s (×R) | GCAD s |
|---|---|---|---|---|---|---|---|---|---|---|
| 98 | finiteBx12_5_279_6_0__2_1_1_0_1_0_0_0_1/symbolic | 2 | monotone+H<br>9025e71 R · ? | 1 | 0.0617 | 0.0213 | 6.47 | 18 | 0.0046 (×0.089) ± | 0.052 |
| 99 | finiteA_4_170_5_0__0_2_0_1_0_1_0_1_0/fixed | 1 | monotone+H<br>9025e71 D · ? | 1 | 0.0617 | 0.0212 | 12.1 | 1 | 0.0021 (×0.086) ± | 0.024 |
| 100 | finiteAx12_4_282_5_0__0_2_0_1_1_0_0_0_1/fixed | 3 | monotone+H<br>9025e71 D · ? | 1 | 0.0617 | 0.0212 | 12.1 | 1 | 0.0017 (×0.083) ± | 0.020 |
| 101 | finiteAx12_4_312_5_0__0_0_0_2_1_1_0_0_1/fixed | 1 | monotone+H<br>9025e71 D · ? | 1 | 0.0617 | 0.0211 | 12.1 | 1 | 0.0017 (×0.097) ± | 0.018 |
| 102 | finiteA_5_314_6_0__0_2_0_1_1_1_0_0_1/fixed | 2 | monotone+H<br>9025e71 D · ? | 1 | 0.0617 | 0.0211 | 12.1 | 1 | 0.0054 (×0.098) ± | 0.055 |
| 103 | finiteA_4_29_5_0__2_0_1_1_1_0_0_0_0/fixed | 1 | monotone+H<br>9025e71 D · ? | 1 | 0.0617 | 0.0212 | 12.1 | 6 | 0.0017 (×0.091) ± | 0.019 |
| 104 | finiteC_5_31_6_0__2_1_1_1_1_0_0_0_0/fixed | 1 | monotone+H<br>9025e71 D · ? | 1 | 0.0617 | 0.0212 | 12.1 | 1 | 0.0041 (×0.098) ± | 0.042 |
| 105 | finiteA_4_29_5_0__2_0_1_1_1_0_0_0_0/symbolic | 1 | monotone+H<br>9025e71 R · ? | 1 | 0.0617 | 0.0212 | 6.05 | 6 | 0.0022 (×0.073) ± | 0.030 |
| 106 | finiteA_3_280_4_0__0_0_0_2_1_0_0_0_1/fixed | 2 | monotone+H<br>9025e71 D · ? | 1 | 0.0617 | 0.0211 | 12.1 | 1 | 0.00079 (×0.082) ± | 0.0096 |
| 107 | finiteAx12_5_313_6_0__2_0_0_1_1_1_0_0_1/fixed | 2 | monotone+H<br>9025e71 D · ? | 1 | 0.0617 | 0.0212 | 12.1 | 1 | 0.0064 (×0.11) ± | 0.056 |
| 108 | finiteA_5_310_6_0__0_2_1_0_1_1_0_0_1/fixed | 1 | monotone+H<br>9025e71 D · ? | 1 | 0.0617 | 0.0213 | 27.6 | 6 | 0.0028 (×0.099) ± | 0.029 |
| 109 | test_ex422_bubble/solve/2 | 1 | optimized+H<br>9025e71 R · ? | 1 | 0.0617 | 0.0211 | 6.04 | 2 | 0.00031 (×0.079) | 0.0039 |
| 110 | finiteC_6_215_11_0__6_1_1_0_1_0_1_1_0/fixed | 4 | monotone+H<br>9025e71 D · ? | 1 | 0.0617 | 0.0212 | 12.1 | 1 | 4.0 (×1.1) ± | 3.5 |
| 111 | finiteE_3_11__3_0__1_1_0_1/fixed | 2 | monotone+H<br>9025e71 D · ? | 1 | 0.0617 | 0.0212 | 12.1 | 1 | 0.0011 (×0.096) ± | 0.011 |
| 112 | finiteA_3_26_4_0__0_2_0_1_1_0_0_0_0/fixed | 1 | monotone+H<br>9025e71 D · ? | 1 | 0.0618 | 0.0212 | 12.1 | 1 | 0.00084 (×0.090) ± | 0.0094 |
| 113 | finiteC_5_412_6_0__0_0_2_1_1_0_0_1_1/fixed | 3 | monotone+H<br>9025e71 D · ? | 1 | 0.0618 | 0.0213 | 12.1 | 1 | 0.0030 (×0.085) ± | 0.035 |
| 114 | finiteAx12_5_314_6_0__0_2_0_1_1_1_0_0_1/fixed | 2 | monotone+H<br>9025e71 D · ? | 1 | 0.0618 | 0.0213 | 12.1 | 1 | 0.0050 (×0.093) ± | 0.054 |
| 115 | finiteCx12_5_158_6_0__0_2_1_1_1_0_0_1_0/fixed | 3 | monotone+H<br>9025e71 D · ? | 1 | 0.0618 | 0.0213 | 12.1 | 1 | 0.0047 (×0.100) ± | 0.047 |
| 116 | finiteA_5_31_6_0__2_1_1_1_1_0_0_0_0/fixed | 1 | monotone+H<br>9025e71 D · ? | 1 | 0.0618 | 0.0211 | 27.3 | 7 | 0.0038 (×0.084) ± | 0.045 |
| 117 | finiteA_5_313_6_0__2_0_0_1_1_1_0_0_1/fixed | 2 | monotone+H<br>9025e71 D · ? | 1 | 0.0618 | 0.0213 | 12.1 | 1 | 0.0062 (×0.11) ± | 0.055 |
| 118 | finiteA_4_154_5_0__0_2_0_1_1_0_0_1_0/fixed | 1 | monotone+H<br>9025e71 D · ? | 1 | 0.0618 | 0.0215 | 12.1 | 1 | 0.0021 (×0.088) ± | 0.024 |
| 119 | test_ex44_T_B2/solve/1 | 1 | optimized<br>9025e71 R · ? | 1 | 0.0619 | 0.0211 | 6.06 | 31 | 0.0045 (×0.22) | 0.021 |
| 120 | finiteCx12_6_231_7_0__2_1_1_0_0_1_1_1_0/fixed | 2 | monotone+H<br>9025e71 D · ? | 1 | 0.0619 | 0.0213 | 26.8 | 1 | 1.3 (×0.87) ± | 1.5 |
| 121 | finiteA_5_313_6_0__2_0_0_1_1_1_0_0_1/symbolic | 2 | monotone+H<br>9025e71 R · ? | 1 | 0.0619 | 0.0215 | 11.4 | 10 | 0.034 (×0.25) ± | 0.14 |
| 122 | finiteA_5_155_6_0__2_1_0_1_1_0_0_1_0/fixed | 1 | monotone+H<br>9025e71 D · ? | 1 | 0.0619 | 0.0213 | 12.1 | 1 | 0.0046 (×0.088) ± | 0.053 |
| 123 | finiteAx12_5_171_6_0__2_1_0_1_0_1_0_1_0/fixed | 1 | monotone+H<br>9025e71 D · ? | 1 | 0.0619 | 0.0212 | 12.1 | 1 | 0.0042 (×0.083) ± | 0.051 |

| # | Original problem / representative full ID | IDs | Method · build | N | Solve s | Check s | RAM MiB | Cells | Matched Rust s (×R) | GCAD s |
|---|---|---|---|---|---|---|---|---|---|---|
| 124 | finiteA_4_284_5_0__0_0_2_1_1_0_0_0_1/fixed | 3 | monotone+H<br>9025e71 D · ? | 1 | 0.0619 | 0.0212 | 12.1 | 5 | 0.0023 **(×0.10)** ± | 0.022 |
| 125 | finiteA_7_446_11_0__0_5_1_1_1_1_0_1_1/fixed | 3 | monotone+H<br>9025e71 D · ? | 1 | 0.0620 | 0.0424 | 28.6 | 8 | — | TO |
| 126 | finiteCx12_6_215_11_0__6_1_1_0_1_0_1_1_0/fixed | 4 | monotone+H<br>9025e71 D · ? | 1 | 0.0620 | 0.0214 | 26.6 | 1 | 4.0 **(×1.1)** ± | 3.6 |
| 127 | finiteE_1_1__3_0__3_0_0_0/fixed | 1 | monotone+H<br>9025e71 D · ? | 1 | 0.0620 | 0.0211 | 16.4 | 1 | 0.00011 **(×0.042)** ± | 0.0026 |
| 128 | test_pentabox_exd/order/1 | 2 | optimized+H<br>9025e71 R · ? | 1 | 0.0621 | 0.0219 | 15.4 | 7 | 0.0077 **(×0.15)** | 0.051 |
| 129 | test_pentabox_exb/solve/3 | 1 | optimized+H<br>9025e71 R · ? | 1 | 0.0621 | 0.0216 | 15.4 | 3 | 0.080 **(×0.40)** | 0.20 |
| 130 | test_pentabox_exb/order/1 | 2 | optimized+H<br>9025e71 R · ? | 1 | 0.0622 | 0.0216 | 15.4 | 7 | 0.082 **(×0.35)** | 0.23 |
| 131 | test_gluza_ex33_trivial/solve/3 | 1 | optimized+H<br>9025e71 R · ? | 1 | 0.0622 | 0.0218 | 19.4 | 1 | — | TO |
| 132 | test_gluza_ex33_trivial/order/1 | 2 | optimized+H<br>9025e71 R · ? | 1 | 0.0622 | 0.0218 | 19.4 | 0 | — | TO |
| 133 | test_pentabox_exd/solve/3 | 1 | optimized+H<br>9025e71 R · ? | 1 | 0.0623 | 0.0218 | 15.4 | 4 | 0.0075 **(×0.16)** | 0.048 |
| 134 | finiteE_2_5__3_0__2_0_1_0/fixed | 1 | monotone+H<br>9025e71 D · ? | 1 | 0.0623 | 0.0212 | 12.1 | 3 | 0.00082 **(×0.14)** ± | 0.0056 |
| 135 | finiteA_5_157_6_0__2_0_1_1_1_0_0_1_0/symbolic | 1 | monotone+H<br>9025e71 R · ? | 1 | 0.0626 | 0.0212 | 6.05 | 6 | 0.0097 **(×0.11)** ± | 0.090 |
| 136 | finiteAx12_5_410_10_0__0_6_0_1_1_0_0_1_1/fixed | 4 | monotone+H<br>9025e71 D · ? | 1 | 0.0627 | 0.0212 | 12.1 | 1 | 0.0084 **(×0.10)** ± | 0.082 |
| 137 | test_ex44_T_C1/solve/1 | 1 | optimized<br>9025e71 R · ? | 1 | 0.0629 | 0.0210 | 6.04 | 3 | 0.021 **(×0.51)** | 0.041 |
| 138 | finiteB_4_85_5_0__2_0_1_0_1_0_1_0_0/fixed | 1 | monotone+H<br>9025e71 D · ? | 1 | 0.0629 | 0.0211 | 12.1 | 3 | 0.0012 **(×0.086)** ± | 0.014 |
| 139 | test_ex44_B1_B2_B4/solve/1 | 1 | optimized<br>9025e71 R · ? | 1 | 0.0629 | 0.0211 | 6.06 | 0 | 0.0058 **(×0.29)** | 0.020 |
| 140 | finiteA_5_155_6_0__2_1_0_1_1_0_0_1_0/symbolic | 1 | monotone+H<br>9025e71 R · ? | 1 | 0.0631 | 0.0215 | 11.4 | 9 | 0.019 **(×0.16)** ± | 0.12 |
| 141 | finiteA_4_153_5_0__2_0_0_1_1_0_0_1_0/symbolic | 1 | monotone+H<br>9025e71 R · ? | 1 | 0.0632 | 0.0211 | 6.05 | 1 | 0.0023 **(×0.090)** ± | 0.026 |
| 142 | finiteA_5_316_10_0__0_0_6_1_1_1_0_0_1/fixed | 5 | monotone+H<br>9025e71 D · ? | 1 | 0.0633 | 0.0213 | 27.7 | 6 | 0.018 **(×0.14)** ± | 0.13 |
| 143 | finiteAx12_5_155_6_0__2_1_0_1_1_0_0_1_0/fixed | 1 | monotone+H<br>9025e71 D · ? | 1 | 0.0635 | 0.0212 | 12.1 | 1 | 0.0047 **(×0.088)** ± | 0.053 |
| 144 | test_ex414_bnp6/solve/2 | 1 | optimized<br>9025e71 R · ? | 1 | 0.0635 | 0.0218 | 15.4 | 2 | 0.00057 **(×0.059)** | 0.0096 |
| 145 | finiteB_4_277_5_0__2_0_1_0_1_0_0_0_1/symbolic | 3 | monotone+H<br>9025e71 R · ? | 1 | 0.0637 | 0.0211 | 6.05 | 3 | 0.0019 **(×0.064)** ± | 0.029 |
| 146 | finiteA_3_296_4_0__0_0_0_2_0_1_0_0_1/symbolic | 2 | monotone+H<br>9025e71 R · ? | 1 | 0.0643 | 0.0244 | 6.05 | 6 | 0.0015 **(×0.092)** ± | 0.016 |
| 147 | test_cube/solve/1 | 1 | optimized<br>9025e71 R · ? | 1 | 0.0644 | 0.0210 | 6.04 | 1 | 0.00020 **(×0.055)** | 0.0037 |
| 148 | test_hypercube_4/solve/1 | 1 | optimized<br>9025e71 R · ? | 1 | 0.0651 | 0.0211 | 9.54 | 1 | 0.00071 **(×0.13)** | 0.0056 |
| 149 | test_ex415_bnp7/solve/2 | 1 | optimized<br>9025e71 R · ? | 1 | 0.0669 | 0.0219 | 19.4 | 10 | 0.0011 **(×0.052)** | 0.020 |

| # | Original problem / representative full ID | IDs | Method · build | N | Solve s | Check s | RAM MiB | Cells | Matched Rust s (×R) | GCAD s |
|---|---|---|---|---|---|---|---|---|---|---|
| 150 | test_boundary/solve/1 | 1 | optimized<br>9025e71 R · ? | 1 | 0.0814 | 0.0210 | 4.40 | 0 | 0.000051 **(×0.038)** | 0.0013 |
| 151 | test_massive_triangle/solve/2 | 1 | optimized<br>9025e71 R · ? | 1 | 0.0815 | 0.0211 | 6.09 | 4 | 0.00079 **(×0.11)** | 0.0071 |
| 152 | test_ex44_B1_B2_B3/solve/1 | 1 | optimized<br>9025e71 R · ? | 1 | 0.0817 | 0.0211 | 12.6 | 4 | 0.013 **(×0.67)** | 0.019 |
| 153 | finiteB_3_21_4_0__2_0_1_0_1_0_0_0_0/symbolic | 1 | monotone+H<br>9025e71 R · ? | 1 | 0.0817 | 0.0212 | 6.06 | 3 | 0.00037 **(×0.042)** ± | 0.0088 |
| 154 | finiteA_5_93_6_0__2_0_1_1_1_0_1_0_0/symbolic | 1 | monotone+H<br>9025e71 R · ? | 1 | 0.0817 | 0.0212 | 14.1 | 58 | 0.013 **(×0.11)** ± | 0.11 |
| 155 | finiteA_4_170_5_0__0_2_0_1_0_1_0_1_0/symbolic | 1 | monotone+H<br>9025e71 R · ? | 1 | 0.0818 | 0.0413 | 14.0 | 151 | 0.0052 **(×0.085)** ± | 0.062 |
| 156 | finiteC_6_95_7_0__2_1_1_1_1_0_1_0_0/fixed | 1 | monotone+H<br>9025e71 D · ? | 1 | 0.0818 | 0.0211 | 12.1 | 1 | 0.021 **(×0.12)** ± | 0.17 |
| 157 | finiteE_2_9__3_0__2_0_0_1/symbolic | 1 | monotone+H<br>9025e71 R · ? | 1 | 0.0819 | 0.0211 | 6.07 | 1 | 0.00084 **(×0.12)** ± | 0.0071 |
| 158 | finiteB_4_277_5_0__2_0_1_0_1_0_0_0_1/fixed | 3 | monotone+H<br>9025e71 D · ? | 1 | 0.0819 | 0.0212 | 26.6 | 3 | 0.0016 **(×0.075)** ± | 0.022 |
| 159 | finiteA_4_85_5_0__2_0_1_0_1_0_1_0_0/fixed | 1 | monotone+H<br>9025e71 D · ? | 1 | 0.0819 | 0.0212 | 26.8 | 15 | 0.0025 **(×0.12)** ± | 0.022 |
| 160 | finiteA_4_89_5_0__2_0_0_1_1_0_1_0_0/fixed | 1 | monotone+H<br>9025e71 D · ? | 1 | 0.0819 | 0.0212 | 26.8 | 8 | 0.0025 **(×0.11)** ± | 0.024 |
| 161 | finiteAx12_5_31_6_0__2_1_1_1_1_0_0_0_0/fixed | 1 | monotone+H<br>9025e71 D · ? | 1 | 0.0819 | 0.0211 | 27.3 | 7 | 0.0036 **(×0.080)** ± | 0.045 |
| 162 | finiteB_5_339_6_0__2_1_0_0_1_0_1_0_1/fixed | 3 | monotone+H<br>9025e71 D · ? | 1 | 0.0819 | 0.0212 | 26.8 | 6 | 0.0034 **(×0.085)** ± | 0.040 |
| 163 | finiteCx12_6_438_7_0__0_2_1_0_1_1_0_1_1/fixed | 1 | monotone+H<br>9025e71 D · ? | 1 | 0.0819 | 0.0213 | 12.1 | 1 | 0.041 **(×0.21)** ± | 0.20 |
| 164 | finiteC_6_470_10_0__0_5_1_0_1_0_1_1_1/fixed | 6 | monotone+H<br>9025e71 D · ? | 1 | 0.0820 | 0.0213 | 27.3 | 7 | 0.14 **(×0.41)** ± | 0.33 |
| 165 | finiteB_5_213_6_0__2_0_1_0_1_0_1_1_0/fixed | 1 | monotone+H<br>9025e71 D · ? | 1 | 0.0820 | 0.0212 | 26.7 | 10 | 0.0025 **(×0.097)** ± | 0.026 |
| 166 | test_melih_box/order/1 | 2 | optimized+H<br>9025e71 R · ? | 1 | 0.0820 | 0.0213 | 6.43 | 1 | 0.0025 **(×0.073)** | 0.035 |
| 167 | finiteC_6_438_7_0__0_2_1_0_1_1_0_1_1/fixed | 1 | monotone+H<br>9025e71 D · ? | 1 | 0.0820 | 0.0213 | 12.1 | 1 | 0.037 **(×0.20)** ± | 0.19 |
| 168 | finiteA_4_344_5_0__0_0_0_2_1_0_1_0_1/fixed | 3 | monotone+H<br>9025e71 D · ? | 1 | 0.0820 | 0.0213 | 26.8 | 10 | 0.0024 **(×0.12)** ± | 0.021 |
| 169 | finiteB_6_343_7_0__2_1_1_0_1_0_1_0_1/fixed | 2 | monotone+H<br>9025e71 D · ? | 1 | 0.0820 | 0.0228 | 27.0 | 15 | 0.0063 **(×0.092)** ± | 0.069 |
| 170 | finiteBx12_5_339_6_0__2_1_0_0_1_0_1_0_1/fixed | 3 | monotone+H<br>9025e71 D · ? | 1 | 0.0820 | 0.0212 | 26.8 | 6 | 0.0034 **(×0.089)** ± | 0.038 |
| 171 | finiteCx12_7_247_10_0__4_1_1_0_1_1_1_1_0/fixed | 3 | monotone+H<br>9025e71 D · ? | 1 | 0.0821 | 0.0212 | 27.1 | 1 | — | TO |
| 172 | finiteAx12_5_301_6_0__2_0_1_1_0_1_0_0_1/fixed | 3 | monotone+H<br>9025e71 D · ? | 1 | 0.0821 | 0.0213 | 26.9 | 8 | 0.0042 **(×0.090)** ± | 0.047 |
| 173 | finiteC_7_247_10_0__4_1_1_0_1_1_1_1_0/fixed | 3 | monotone+H<br>9025e71 D · ? | 1 | 0.0821 | 0.0214 | 27.1 | 1 | — | TO |
| 174 | finiteAx12_5_316_10_0__0_0_6_1_1_1_0_0_1/fixed | 5 | monotone+H<br>9025e71 D · ? | 1 | 0.0821 | 0.0213 | 27.5 | 6 | 0.011 **(×0.15)** ± | 0.073 |
| 175 | finiteAx12_5_316_10_0__0_0_6_1_1_1_0_0_1/symbolic | 5 | monotone+H<br>9025e71 R · ? | 1 | 0.0821 | 0.0424 | 14.6 | 249 | — | TO |

| # | Original problem / representative full ID | IDs | Method · build | N | Solve s | Check s | RAM MiB | Cells | Matched Rust s (×R) | GCAD s |
|---|---|---|---|---|---|---|---|---|---|---|
| 176 | finiteA_5_412_6_0__0_0_2_1_1_0_0_1_1/fixed | 3 | monotone+H<br>9025e71 D · ? | 1 | 0.0821 | 0.0212 | 27.5 | 6 | 0.014 **(×0.16)** ± | 0.089 |
| 177 | finiteA_6_190_7_0__0_2_1_1_1_1_0_1_0/fixed | 1 | monotone+H<br>9025e71 D · ? | 1 | 0.0822 | 0.0213 | 26.7 | 1 | 0.0095 **(×0.087)** ± | 0.11 |
| 178 | finiteB_5_341_6_0__2_0_1_0_1_0_1_0_1/fixed | 3 | monotone+H<br>9025e71 D · ? | 1 | 0.0822 | 0.0212 | 26.8 | 11 | 0.0057 **(×0.11)** ± | 0.051 |
| 179 | finiteA_5_316_10_0__0_0_6_1_1_1_0_0_1/symbolic | 5 | monotone+H<br>9025e71 R · ? | 1 | 0.0822 | 0.0416 | 14.7 | 151 | — | TO |
| 180 | finiteA_5_314_6_0__0_2_0_1_1_1_0_0_1/symbolic | 2 | monotone+H<br>9025e71 R · ? | 1 | 0.0822 | 0.0431 | 14.4 | 144 | 0.54 **(×0.68)** ± | 0.80 |
| 181 | finiteBx12_6_343_7_0__2_1_1_0_1_0_1_0_1/fixed | 2 | monotone+H<br>9025e71 D · ? | 1 | 0.0823 | 0.0212 | 27.1 | 15 | 0.0065 **(×0.092)** ± | 0.070 |
| 182 | finiteA_6_414_10_0__0_5_1_1_1_0_0_1_1/fixed | 4 | monotone+H<br>9025e71 D · ? | 1 | 0.0823 | 0.0213 | 28.4 | 7 | 3.3 **(×1.0)** ± | 3.1 |
| 183 | finiteA_5_376_6_0__0_0_0_2_1_1_1_0_1/fixed | 3 | monotone+H<br>9025e71 D · ? | 1 | 0.0823 | 0.0212 | 26.9 | 12 | 0.0061 **(×0.13)** ± | 0.048 |
| 184 | test_ex412_box/solve/2 | 1 | optimized<br>9025e71 R · ? | 1 | 0.0824 | 0.0213 | 6.43 | 1 | 0.00023 **(×0.035)** | 0.0065 |
| 185 | finiteB_5_453_6_0__2_0_1_0_0_0_1_1_1/symbolic | 1 | monotone+H<br>9025e71 R · ? | 1 | 0.0824 | 0.0615 | 15.0 | 524 | 0.0045 **(×0.094)** ± | 0.048 |
| 186 | finiteA_5_309_6_0__2_0_1_0_1_1_0_0_1/fixed | 2 | monotone+H<br>9025e71 D · ? | 1 | 0.0824 | 0.0421 | 27.9 | 42 | 0.018 **(×0.30)** ± | 0.058 |
| 187 | test_ex413_pentagon/solve/2 | 1 | optimized<br>9025e71 R · ? | 1 | 0.0825 | 0.0219 | 19.0 | 3 | 0.00052 **(×0.037)** | 0.014 |
| 188 | finiteAx12_5_376_6_0__0_0_0_2_1_1_1_0_1/fixed | 3 | monotone+H<br>9025e71 D · ? | 1 | 0.0825 | 0.0212 | 26.8 | 12 | 0.0058 **(×0.12)** ± | 0.048 |
| 189 | finiteA_5_301_6_0__2_0_1_1_0_1_0_0_1/fixed | 3 | monotone+H<br>9025e71 D · ? | 1 | 0.0826 | 0.0211 | 26.9 | 8 | 0.0046 **(×0.095)** ± | 0.048 |
| 190 | finiteA_5_121_6_0__2_0_0_1_1_1_1_0_0/fixed | 1 | monotone+H<br>9025e71 D · ? | 1 | 0.0826 | 0.0211 | 27.1 | 9 | 0.0036 **(×0.086)** ± | 0.042 |
| 191 | test_pentabox_exb_symb/order/1 | 1 | optimized<br>9025e71 R · ? | 1 | 0.0830 | 0.0221 | 23.0 | 5 | — | TO |
| 192 | finiteA_2_17_3_0__2_0_0_0_1_0_0_0_0/symbolic | 1 | monotone+H<br>9025e71 R · ? | 1 | 0.0831 | 0.0211 | 6.07 | 1 | 0.00019 **(×0.032)** ± | 0.0058 |
| 193 | finiteA_6_159_7_0__2_1_1_1_1_0_0_1_0/fixed | 1 | monotone+H<br>9025e71 D · ? | 1 | 0.0832 | 0.0212 | 27.5 | 7 | 0.029 **(×0.10)** ± | 0.28 |
| 194 | finiteA_5_87_6_0__2_1_1_0_1_0_1_0_0/fixed | 1 | monotone+H<br>9025e71 D · ? | 1 | 0.0832 | 0.0413 | 27.1 | 28 | 0.0045 **(×0.11)** ± | 0.041 |
| 195 | finiteE_2_5__3_0__2_0_1_0/symbolic | 1 | monotone+H<br>9025e71 R · ? | 1 | 0.0833 | 0.0212 | 6.07 | 3 | 0.00052 **(×0.073)** ± | 0.0071 |
| 196 | finiteC_6_476_10_0__0_0_5_1_1_0_1_1_1/fixed | 4 | monotone+H<br>9025e71 D · ? | 1 | 0.0834 | 0.0211 | 27.5 | 7 | 0.038 **(×0.19)** ± | 0.20 |
| 197 | finiteA_5_157_6_0__2_0_1_1_1_0_0_1_0/fixed | 1 | monotone+H<br>9025e71 D · ? | 1 | 0.0836 | 0.0212 | 27.2 | 6 | 0.0049 **(×0.087)** ± | 0.057 |
| 198 | finiteB_4_197_5_0__2_0_1_0_0_0_1_1_0/fixed | 1 | monotone+H<br>9025e71 D · ? | 1 | 0.0836 | 0.0212 | 27.1 | 13 | 0.0012 **(×0.097)** ± | 0.013 |
| 199 | finiteA_6_318_7_0__0_2_1_1_1_1_0_0_1/fixed | 2 | monotone+H<br>9025e71 D · ? | 1 | 0.0837 | 0.0213 | 28.5 | 7 | 0.72 **(×0.50)** ± | 1.5 |
| 200 | finiteAx12_4_170_5_0__0_2_0_1_0_1_0_1_0/symbolic | 1 | monotone+H<br>9025e71 R · ? | 1 | 0.0837 | 0.0413 | 11.0 | 151 | 0.0036 **(×0.094)** ± | 0.038 |
| 201 | finiteAx12_6_159_7_0__2_1_1_1_1_0_0_1_0/fixed | 1 | monotone+H<br>9025e71 D · ? | 1 | 0.0837 | 0.0211 | 27.3 | 7 | 0.020 **(×0.13)** ± | 0.16 |

| # | Original problem / representative full ID | IDs | Method · build | N | Solve s | Check s | RAM MiB | Cells | Matched Rust s (×R) | GCAD s |
|---|---|---|---|---|---|---|---|---|---|---|
| 202 | finiteC_7_478_10_0__0_4_1_1_1_0_1_1_1/fixed | 5 | monotone+H<br>9025e71 D · ? | 1 | 0.0838 | 0.0416 | 28.1 | 8 | — | TO |
| 203 | finiteAx12_5_286_6_0__0_2_1_1_1_0_0_0_1/fixed | 3 | monotone+H<br>9025e71 D · ? | 1 | 0.0839 | 0.0212 | 27.4 | 6 | 0.0043 **(×0.087)** ± | 0.049 |
| 204 | test_massive_triangle/solve/1 | 1 | optimized<br>9025e71 R · ? | 1 | 0.0846 | 0.0212 | 6.05 | 6 | 0.0010 **(×0.13)** | 0.0076 |
| 205 | finiteA_4_293_5_0__2_0_1_0_0_1_0_0_1/fixed | 3 | monotone+H<br>9025e71 D · ? | 1 | 0.0846 | 0.0212 | 27.3 | 6 | 0.0016 **(×0.11)** ± | 0.015 |
| 206 | finiteAx12_6_318_7_0__0_2_1_1_1_1_0_0_1/fixed | 2 | monotone+H<br>9025e71 D · ? | 1 | 0.0848 | 0.0213 | 27.8 | 7 | 0.67 **(×0.50)** ± | 1.4 |
| 207 | finiteAx12_5_313_6_0__2_0_0_1_1_1_0_0_1/symbolic | 2 | monotone+H<br>9025e71 R · ? | 1 | 0.0895 | 0.0218 | 13.8 | 10 | 0.032 **(×0.28)** ± | 0.11 |
| 208 | test_hypercube_1/solve/1 | 1 | optimized<br>9025e71 R · ? | 1 | 0.102 | 0.0211 | 6.03 | 1 | 0.00013 **(×0.035)** | 0.0037 |
| 209 | finiteAx12_7_381_10_0__4_0_1_1_1_1_1_0_1/fixed | 4 | monotone+H<br>0a95bc2 R · C | 3 | 0.102 | 0.0614 | 18.6 | 836 | — | TO |
| 210 | finiteA_5_412_6_0__0_0_2_1_1_0_0_1_1/symbolic | 3 | monotone+H<br>9025e71 R · ? | 1 | 0.102 | 0.0213 | 6.49 | 6 | 0.11 **(×0.22)** ± | 0.49 |
| 211 | finiteCx12_6_476_10_0__0_0_5_1_1_0_1_1_1/fixed | 4 | monotone+H<br>9025e71 D · ? | 1 | 0.102 | 0.0212 | 27.3 | 7 | 0.040 **(×0.19)** ± | 0.21 |
| 212 | finiteAx12_6_414_10_0__0_5_1_1_1_0_0_1_1/fixed | 4 | monotone+H<br>9025e71 D · ? | 1 | 0.102 | 0.0213 | 28.5 | 7 | 3.1 **(×1.0)** ± | 3.0 |
| 213 | finiteAx12_6_190_7_0__0_2_1_1_1_1_0_1_0/fixed | 1 | monotone+H<br>9025e71 D · ? | 1 | 0.102 | 0.0212 | 26.7 | 1 | 0.0094 **(×0.085)** ± | 0.11 |
| 214 | finiteC_6_231_7_0__2_1_1_0_0_1_1_1_0/fixed | 2 | monotone+H<br>9025e71 D · ? | 1 | 0.102 | 0.0212 | 26.7 | 1 | 1.2 **(×0.84)** ± | 1.5 |
| 215 | finiteA_6_125_7_0__2_0_1_1_1_1_1_0_0/symbolic | 1 | monotone+H<br>9025e71 R · ? | 1 | 0.102 | 0.0417 | 15.4 | 67 | 0.029 **(×0.13)** ± | 0.22 |
| 216 | finiteA_5_93_6_0__2_0_1_1_1_0_1_0_0/fixed | 1 | monotone+H<br>9025e71 D · ? | 1 | 0.103 | 0.0624 | 27.7 | 58 | 0.0078 **(×0.14)** ± | 0.058 |
| 217 | finiteB_5_453_6_0__2_0_1_0_0_0_1_1_1/fixed | 1 | monotone+H<br>9025e71 D · ? | 1 | 0.103 | 0.0419 | 27.6 | 77 | 0.0041 **(×0.12)** ± | 0.035 |
| 218 | finiteE_4_15__4_0__1_1_1_1/symbolic | 2 | monotone+H<br>9025e71 R · ? | 1 | 0.103 | 0.0618 | 16.4 | 406 | 0.013 **(×0.13)** ± | 0.10 |
| 219 | test_ex44_B1_B2/solve/1 | 1 | optimized<br>9025e71 R · ? | 1 | 0.104 | 0.0211 | 6.04 | 1 | 0.0015 **(×0.18)** | 0.0082 |
| 220 | test_ex44_B1_C1/solve/1 | 1 | optimized<br>9025e71 R · ? | 1 | 0.107 | 0.0226 | 6.03 | 2 | 0.0051 **(×0.39)** | 0.013 |
| 221 | finiteAx12_5_155_6_0__2_1_0_1_1_0_0_1_0/symbolic | 1 | monotone+H<br>9025e71 R · ? | 1 | 0.107 | 0.0224 | 13.8 | 9 | 0.016 **(×0.18)** ± | 0.089 |
| 222 | finiteD_6_219_7_0__2_1_0_1_1_0_1_1_0/symbolic | 1 | monotone+H<br>9025e71 R · ? | 1 | 0.113 | 0.0417 | 16.3 | 262 | 4.5 **(×0.83)** ± | 5.4 |
| 223 | test_ex44_HT_HB2_HB3/solve/1 | 1 | optimized<br>9025e71 R · ? | 1 | 0.123 | 0.102 | 13.5 | 0 | 0.058 **(×0.48)** | 0.12 |
| 224 | finiteC_5_412_6_0__0_0_2_1_1_0_0_1_1/symbolic | 3 | monotone+H<br>9025e71 R · ? | 1 | 0.123 | 0.0830 | 15.1 | 228 | 0.028 **(×0.28)** ± | 0.10 |
| 225 | test_ex415_bnp7/solve/1 | 1 | optimized<br>9025e71 R · ? | 1 | 0.123 | 0.0220 | 19.4 | 18 | 0.0010 **(×0.049)** | 0.021 |
| 226 | finiteA_6_317_7_0__2_0_1_1_1_1_0_0_1/fixed | 2 | monotone+H<br>9025e71 D · ? | 1 | 0.123 | 0.0414 | 28.9 | 56 | 0.15 **(×0.43)** ± | 0.36 |
| 227 | finiteAx12_6_190_7_0__0_2_1_1_1_1_0_1_0/symbolic | 1 | monotone+H<br>9025e71 R · ? | 1 | 0.124 | 0.0422 | 19.4 | 130 | 0.055 **(×0.24)** ± | 0.23 |

| # | Original problem / representative full ID | IDs | Method · build | N | Solve s | Check s | RAM MiB | Cells | Matched Rust s (×R) | GCAD s |
|---|---|---|---|---|---|---|---|---|---|---|
| 228 | finiteCx12_7_478_10_0__0_4_1_1_1_0_1_1_1/fixed | 5 | monotone+H<br>9025e71 D · ? | 1 | 0.124 | 0.0414 | 28.6 | 8 | — | TO |
| 229 | finiteAx12_5_410_10_0__0_6_0_1_1_0_0_1_1/symbolic | 4 | monotone+H<br>9025e71 R · ? | 1 | 0.125 | 0.143 | 25.4 | 973 | 0.30 **(×0.35)** ± | 0.85 |
| 230 | failing_test_ex44_T_C1_B2/solve/1 | 1 | optimized<br>9025e71 R · ? | 1 | 0.125 | 0.0843 | 13.3 | 57 | 0.20 **(×1.3)** | 0.16 |
| 231 | finiteAx12_6_317_7_0__2_0_1_1_1_1_0_0_1/fixed | 2 | monotone+H<br>9025e71 D · ? | 1 | 0.125 | 0.0416 | 28.9 | 56 | 0.14 **(×0.39)** ± | 0.35 |
| 232 | finiteAx12_5_314_6_0__0_2_0_1_1_1_0_0_1/symbolic | 2 | monotone+H<br>9025e71 R · ? | 1 | 0.131 | 0.0888 | 15.5 | 144 | 0.49 **(×0.75)** ± | 0.65 |
| 233 | finiteAx12_7_446_11_0__0_5_1_1_1_1_0_1_1/fixed | 3 | monotone+H<br>9025e71 D · ? | 1 | 0.142 | 0.0416 | 28.5 | 8 | — | TO |
| 234 | test_ex414_bnp6/solve/1 | 1 | optimized<br>9025e71 R · ? | 1 | 0.143 | 0.0219 | 15.4 | 1 | 0.00039 **(×0.040)** | 0.0099 |
| 235 | finiteF_4_15__4_0__1_1_1_1/symbolic | 1 | monotone+H<br>9025e71 R · ? | 1 | 0.143 | 0.144 | 19.4 | 624 | 0.071 **(×0.47)** ± | 0.15 |
| 236 | finiteC_5_227_6_0__2_1_0_0_0_1_1_1_0/symbolic | 3 | monotone+H<br>9025e71 R · ? | 1 | 0.144 | 0.163 | 20.2 | 632 | 12 **(×0.92)** ± | 13 |
| 237 | failing_test_ex43/project/1 | 2 | optimized<br>9025e71 R · ? | 1 | 0.158 | 0.0211 | 12.6 | 10 | 0.079 **(×0.29)** | 0.28 |
| 238 | test_ex44_HB1_HB2_HB3/solve/1 | 1 | optimized<br>9025e71 R · ? | 1 | 0.163 | 0.147 | 14.1 | 0 | 0.033 **(×0.69)** | 0.048 |
| 239 | test_ex413_pentagon/solve/1 | 1 | optimized<br>9025e71 R · ? | 1 | 0.163 | 0.0219 | 19.0 | 5 | 0.00052 **(×0.037)** | 0.014 |
| 240 | finiteBx12_5_339_6_0__2_1_0_0_1_0_1_0_1/symbolic | 3 | monotone+H<br>9025e71 R · ? | 1 | 0.164 | 0.123 | 25.0 | 954 | 0.027 **(×0.16)** ± | 0.17 |
| 241 | test_gluza_ex33/order/1 | 2 | linear_negative<br>256527b R · C | 3 | 0.164 | 0.0209 | 11.4 | 3 | — | TO |
| 242 | finiteE_3_13__3_0__1_0_1_1/symbolic | 1 | monotone+H<br>9025e71 R · ? | 1 | 0.166 | 0.0212 | 6.07 | 4 | 0.0020 **(×0.069)** ± | 0.028 |
| 243 | finiteD_6_219_7_0__2_1_0_1_1_0_1_1_0/fixed | 1 | monotone+H<br>9025e71 D · ? | 1 | 0.166 | 0.123 | 29.7 | 262 | 0.061 **(×0.46)** ± | 0.13 |
| 244 | test_massless_planar_elliptic_symb/order/1 | 1 | local+H<br>e43a2c4 R · C | 1 | 0.183 | 0.0211 | 6.00 | 12 | — | TO |
| 245 | finiteBx12_6_343_7_0__2_1_1_0_1_0_1_0_1/symbolic | 2 | monotone+H<br>9025e71 R · ? | 1 | 0.184 | 0.225 | 44.7 | 1963 | 0.075 **(×0.23)** ± | 0.33 |
| 246 | finiteA_6_125_7_0__2_0_1_1_1_1_1_0_0/fixed | 1 | monotone+H<br>9025e71 D · ? | 1 | 0.184 | 0.103 | 29.1 | 67 | 0.014 **(×0.12)** ± | 0.12 |
| 247 | test_gluza_ex30/order/1 | 2 | linear_negative<br>256527b R · C | 3 | 0.185 | 0.0210 | 6.03 | 1 | — | TO |
| 248 | finiteB_6_469_7_0__2_0_1_0_1_0_1_1_1/fixed | 1 | monotone+H<br>9025e71 D · ? | 1 | 0.186 | 0.102 | 29.4 | 184 | 0.031 **(×0.22)** ± | 0.14 |
| 249 | finiteA_7_381_10_0__4_0_1_1_1_1_1_0_1/fixed | 4 | monotone+H<br>0a95bc2 R · C | 3 | 0.186 | 0.0613 | 16.2 | 836 | — | TO |
| 250 | test_pentabox_exd_symb/order/1 | 1 | optimized<br>9025e71 R · ? | 1 | 0.204 | 0.0425 | 23.0 | 152 | — | TO |
| 251 | finiteA_5_410_10_0__0_6_0_1_1_0_0_1_1/symbolic | 4 | monotone+H<br>9025e71 R · ? | 1 | 0.204 | 0.164 | 22.8 | 973 | 0.77 **(×0.35)** ± | 2.2 |
| 252 | finiteB_6_343_7_0__2_1_1_0_1_0_1_0_1/symbolic | 2 | monotone+H<br>9025e71 R · ? | 1 | 0.205 | 0.285 | 46.5 | 2249 | 0.074 **(×0.24)** ± | 0.30 |
| 253 | finiteC_5_214_10_0__0_6_1_0_1_0_1_1_0/symbolic | 5 | monotone+H<br>9025e71 R · ? | 1 | 0.206 | 0.164 | 20.2 | 632 | — | TO |

| # | Original problem / representative full ID | IDs | Method · build | N | Solve s | Check s | RAM MiB | Cells | Matched Rust s (×R) | GCAD s |
|---|---|---|---|---|---|---|---|---|---|---|
| 254 | finiteA_3_280_4_0__0_0_0_2_1_0_0_0_1/symbolic | 2 | monotone+H<br>9025e71 R · ? | 1 | 0.223 | 0.0212 | 6.05 | 1 | 0.0010 **(×0.094)** ± | 0.011 |
| 255 | finiteBx12_5_279_6_0__2_1_1_0_1_0_0_0_1/fixed | 2 | monotone+H<br>9025e71 D · ? | 1 | 0.223 | 0.0211 | 26.7 | 4 | 0.0024 **(×0.078)** ± | 0.030 |
| 256 | finiteCx12_5_158_6_0__0_2_1_1_1_0_0_1_0/symbolic | 3 | optimized<br>d9488c7 R · ? | 1 | 0.224 | 0.245 | 28.5 | 1273 | 0.74 **(×0.73)** ± | 1.0 |
| 257 | finiteA_6_190_7_0__0_2_1_1_1_1_0_1_0/symbolic | 1 | monotone+H<br>9025e71 R · ? | 1 | 0.224 | 0.0648 | 19.4 | 130 | 0.098 **(×0.29)** ± | 0.33 |
| 258 | finiteD_7_375_13_0__7_1_1_0_1_1_1_0_1/fixed | 4 | monotone+H<br>9025e71 D · ? | 1 | 0.224 | 0.124 | 31.1 | 296 | 3.3 **(×0.81)** ± | 4.0 |
| 259 | finiteD_6_373_7_0__2_0_1_0_1_1_1_0_1/fixed | 2 | monotone+H<br>9025e71 D · ? | 1 | 0.225 | 0.123 | 30.3 | 214 | 0.83 **(×0.71)** ± | 1.2 |
| 260 | finiteBx12_7_471_13_0__7_1_1_0_1_0_1_1_1/fixed | 4 | monotone+H<br>9025e71 D · ? | 1 | 0.226 | 0.164 | 31.0 | 245 | 0.067 **(×0.28)** ± | 0.24 |
| 261 | finiteA_5_345_6_0__2_0_0_1_1_0_1_0_1/fixed | 4 | monotone+H<br>9025e71 D · ? | 1 | 0.226 | 0.185 | 29.7 | 309 | 0.033 **(×0.32)** ± | 0.10 |
| 262 | finiteB_7_471_13_0__7_1_1_0_1_0_1_1_1/fixed | 4 | monotone+H<br>9025e71 D · ? | 1 | 0.227 | 0.164 | 31.0 | 245 | 0.057 **(×0.25)** ± | 0.23 |
| 263 | finiteD_7_251_12_0__6_1_0_1_1_1_1_1_0/fixed | 6 | monotone+H<br>9025e71 D · ? | 1 | 0.227 | 0.246 | 31.5 | 364 | — | partial TO |
| 264 | finiteB_6_469_7_0__2_0_1_0_1_0_1_1_1/symbolic | 1 | monotone+H<br>9025e71 R · ? | 1 | 0.232 | 0.0417 | 14.8 | 184 | 0.60 **(×0.54)** ± | 1.1 |
| 265 | finiteB_5_279_6_0__2_1_1_0_1_0_0_0_1/fixed | 2 | monotone+H<br>9025e71 D · ? | 1 | 0.244 | 0.0211 | 26.7 | 4 | 0.0023 **(×0.074)** ± | 0.032 |
| 266 | test_elliptic2l_physical/order/1 | 1 | optimized+H<br>9025e71 R · ? | 1 | 0.246 | 0.709 | 27.1 | 1196 | 1.8 **(×0.89)** | 2.0 |
| 267 | test_elliptic2l_physical/order/2 | 1 | optimized+H<br>9025e71 R · ? | 1 | 0.246 | 0.690 | 27.1 | 1196 | 1.2 **(×0.75)** | 1.6 |
| 268 | test_ex44_HT_C1_HB2/solve/1 | 1 | optimized<br>9025e71 R · ? | 1 | 0.247 | 0.366 | 19.7 | 0 | 0.16 **(×0.71)** | 0.23 |
| 269 | test_unbounded/solve/1 | 1 | optimized+H<br>9025e71 R · ? | 1 | 0.247 | 0.0210 | 8.81 | 1 | 0.000056 **(×0.053)** | 0.0010 |
| 270 | finiteAx12_5_376_6_0__0_0_0_2_1_1_1_0_1/symbolic | 3 | monotone+H<br>9025e71 R · ? | 1 | 0.265 | 0.326 | 39.9 | 1744 | 21 **(×0.93)** ± | 23 |
| 271 | finiteD_6_373_7_0__2_0_1_0_1_1_1_0_1/symbolic | 2 | monotone+H<br>9025e71 R · ? | 1 | 0.265 | 0.325 | 58.9 | 3096 | — | TO |
| 272 | finiteA_5_286_6_0__0_2_1_1_1_0_0_0_1/symbolic | 3 | optimized<br>d9488c7 R · ? | 1 | 0.279 | 0.427 | 44.4 | 3709 | 1.6 **(×0.61)** ± | 2.7 |
| 273 | finiteC_5_158_6_0__0_2_1_1_1_0_0_1_0/symbolic | 3 | optimized<br>d9488c7 R · ? | 1 | 0.286 | 0.447 | 45.6 | 2197 | 1.2 **(×0.87)** ± | 1.3 |
| 274 | finiteB_5_339_6_0__2_1_0_0_1_0_1_0_1/symbolic | 3 | monotone+H<br>9025e71 R · ? | 1 | 0.306 | 0.163 | 27.5 | 1079 | 0.11 **(×0.41)** ± | 0.26 |
| 275 | finiteA_6_377_10_0__5_0_0_1_1_1_1_0_1/fixed | 4 | monotone+H<br>9025e71 D · ? | 1 | 0.307 | 0.225 | 31.5 | 413 | 1.1 **(×0.93)** ± | 1.2 |
| 276 | finiteAx12_6_377_10_0__5_0_0_1_1_1_1_0_1/fixed | 4 | monotone+H<br>9025e71 D · ? | 1 | 0.329 | 0.225 | 31.5 | 413 | 0.92 **(×0.88)** ± | 1.0 |
| 277 | finiteA_5_345_6_0__2_0_0_1_1_0_1_0_1/symbolic | 4 | monotone+H<br>9025e71 R · ? | 1 | 0.348 | 0.548 | 64.3 | 3706 | 0.38 **(×0.65)** ± | 0.59 |
| 278 | finiteAx12_5_286_6_0__0_2_1_1_1_0_0_0_1/symbolic | 3 | optimized<br>d9488c7 R · ? | 1 | 0.372 | 0.588 | 59.9 | 5489 | 3.1 **(×0.72)** ± | 4.3 |
| 279 | test_elliptic2l_physical/solve/3 | 1 | optimized+H<br>9025e71 R · ? | 1 | 0.468 | 0.770 | 39.5 | 1193 | 0.075 **(×0.49)** | 0.15 |

| # | Original problem / representative full ID | IDs | Method · build | N | Solve s | Check s | RAM MiB | Cells | Matched Rust s (×R) | GCAD s |
|---|---|---|---|---|---|---|---|---|---|---|
| 280 | finiteAx12_5_301_6_0__2_0_1_1_0_1_0_0_1/symbolic | 3 | monotone+H<br>9025e71 R · ? | 1 | 0.472 | 0.712 | 103 | 5462 | 3.1 (×0.76) ± | 4.0 |
| 281 | test_elliptic2l_physical/solve/4 | 1 | optimized+H<br>9025e71 R · ? | 1 | 0.530 | 0.811 | 56.4 | 3254 | 0.078 (×0.47) | 0.16 |
| 282 | finiteA_5_301_6_0__2_0_1_1_0_1_0_0_1/symbolic | 3 | monotone+H<br>9025e71 R · ? | 1 | 0.592 | 1.02 | 145 | 7929 | 4.5 (×0.95) ± | 4.8 |
| 283 | finiteAx12_5_171_6_0__2_1_0_1_0_1_0_1_0/symbolic | 1 | monotone+H<br>9025e71 R · ? | 1 | 1.18 | 2.14 | 252 | 15730 | 0.021 (×0.19) ± | 0.11 |
| 284 | test_elliptic2l_physical_symb/solve/2 | 1 | optimized+H<br>9025e71 R · ? | 1 | 1.36 | 2.55 | 196 | 5831 | — | TO |
| 285 | finiteA_5_376_6_0__0_0_0_2_1_1_1_0_1/symbolic | 3 | monotone+H<br>9025e71 R · ? | 1 | 1.68 | 2.38 | 225 | 12357 | 27 (×0.95) ± | 28 |
| 286 | finiteA_5_171_6_0__2_1_0_1_0_1_0_1_0/symbolic | 1 | monotone+H<br>9025e71 R · ? | 1 | 1.81 | 2.63 | 252 | 15730 | 0.031 (×0.19) ± | 0.16 |
| 287 | finiteAx12_5_31_6_0__2_1_1_1_1_0_0_0_0/symbolic | 1 | monotone+H<br>9025e71 R · ? | 1 | 2.32 | 4.15 | 590 | 35354 | 2.8 (×0.68) ± | 4.2 |
| 288 | finiteAx12_6_159_7_0__2_1_1_1_1_0_0_1_0/symbolic | 1 | monotone+H<br>9025e71 R · ? | 1 | 2.85 | 5.26 | 635 | 35354 | — | TO |
| 289 | test_ex_e5/order/1 | 2 | optimized+H<br>9025e71 R · ? | 1 | 4.95 | 11.2 | 170 | 0 | 9.3 (×2.7) | 3.4 |
| 290 | finiteA_5_31_6_0__2_1_1_1_1_0_0_0_0/symbolic | 1 | monotone+H<br>9025e71 R · ? | 1 | 4.98 | 8.82 | 1235 | 74351 | 1.5 (×0.58) ± | 2.6 |
| 291 | test_ex_e5/solve/3 | 1 | optimized+H<br>9025e71 R · ? | 1 | 5.11 | 11.4 | 257 | 13266 | 9.4 (×2.1) | 4.5 |
| 292 | test_ex_e5/solve/4 | 1 | optimized+H<br>9025e71 R · ? | 1 | 5.34 | 14.0 | 373 | 16232 | 10 (×1.9) | 5.2 |
| 293 | test_ex_e5/solve/5 | 1 | optimized+H<br>9025e71 R · ? | 1 | 5.84 | 15.0 | 646 | 54026 | 10 (×1.2) | 9.0 |
| 294 | finiteA_6_159_7_0__2_1_1_1_1_0_0_1_0/symbolic | 1 | monotone+H<br>9025e71 R · ? | 1 | 6.26 | 11.7 | 1329 | 74351 | — | TO |
| 295 | test_massless_planar_elliptic/order/1 | 2 | baseline<br>1d15d03 R · S | 1 | 9.36 | 23.4 | 1630 | 131072 | 9.9 (×0.62) | 16 |
| 296 | test_massless_planar_elliptic/solve/3 | 1 | baseline<br>1d15d03 R · S | 1 | 9.37 | 23.6 | 1647 | 138678 | 9.8 (×0.62) | 16 |
| 297 | finiteC_6_438_7_0__0_2_1_0_1_1_0_1_1/symbolic | 1 | monotone+H<br>9025e71 R · ? | 1 | 13.5 | 26.3 | 2357 | 123552 | 30 (×0.93) ± | 32 |
| 298 | finiteCx12_6_438_7_0__0_2_1_0_1_1_0_1_1/symbolic | 1 | monotone+H<br>9025e71 R · ? | 1 | 21.4 | 32.7 | 2358 | 123552 | 44 (×0.93) ± | 48 |
| 299 | test_gluza_ex30/solve/3 | 1 | bilinear_positive<br>ff22294 R · C | 3 | 22.6 | 16.9 | 744 | 80980 | — | TO |
| 300 | test_elliptic2l_physical_symb/order/1 | 1 | monotone−+H<br>1e29f04 R · C | 1 | 33.9 | 73.1 | 6473 | 269061 | — | TO |
| 301 | finiteC_5_31_6_0__2_1_1_1_1_0_0_0_0/symbolic | 1 | monotone+H<br>9025e71 R · ? | 1 | 35.2 | 124 | 2286† | 271030 | 1.1 (×0.83) ± | 1.3 |
| 302 | finiteC_6_95_7_0__2_1_1_1_1_0_1_0_0/symbolic | 1 | monotone+H<br>9025e71 R · ? | 1 | 44.0 | 146 | 2527† | 271030 | — | TO |
| 303 | hanging_test_ex38/solve/1 | 1 | optimized<br>9025e71 R · ? | 1 | 56.8 | 80.8 | 805† | 129064 | 55 (×3.1) | 18 |
| 304 | finiteA_6_317_7_0__2_0_1_1_1_1_0_0_1/symbolic | 2 | **slow / incomplete** | — | TO 3600 | — | 1364 | — | — | TO |
| 305 | finiteA_6_318_7_0__0_2_1_1_1_1_0_0_1/symbolic | 2 | **slow / incomplete** | — | TO 3600 | — | 1413 | — | — | TO |
| 306 | finiteA_6_377_10_0__5_0_0_1_1_1_1_0_1/symbolic | 4 | **slow / incomplete** | — | TO 3600 | — | 415 | — | — | TO |
| 307 | finiteA_6_414_10_0__0_5_1_1_1_0_0_1_1/symbolic | 4 | **slow / incomplete** | — | TO 3600 | — | 970 | — | — | TO |

| # | Original problem / representative full ID | IDs | Method · build | N | Solve s | Check s | RAM MiB | Cells | Matched Rust s (×R) | GCAD s |
|---|---|---|---|---|---|---|---|---|---|---|
| 308 | finiteA_7_381_10_0__4_0_1_1_1_1_1_0_1/symbolic | 4 | **slow / incomplete** | — | TO 3600 | — | 467 | — | — | TO |
| 309 | finiteA_7_446_11_0__0_5_1_1_1_1_0_1_1/symbolic | 3 | **slow / incomplete** | — | TO 3600 | — | 1944 | — | — | TO |
| 310 | finiteAx12_6_317_7_0__2_0_1_1_1_1_0_0_1/symbolic | 2 | **slow / incomplete** | — | TO 3600 | — | 1430 | — | — | TO |
| 311 | finiteAx12_6_318_7_0__0_2_1_1_1_1_0_0_1/symbolic | 2 | **slow / incomplete** | — | TO 3600 | — | 548 | — | — | TO |
| 312 | finiteAx12_6_377_10_0__5_0_0_1_1_1_1_0_1/symbolic | 4 | **slow / incomplete** | — | TO 3600 | — | 2338 | — | — | TO |
| 313 | finiteAx12_6_414_10_0__0_5_1_1_1_0_0_1_1/symbolic | 4 | **slow / incomplete** | — | TO 3600 | — | 608 | — | — | TO |
| 314 | finiteAx12_7_381_10_0__4_0_1_1_1_1_1_0_1/symbolic | 4 | **slow / incomplete** | — | TO 3600 | — | 485 | — | — | TO |
| 315 | finiteAx12_7_446_11_0__0_5_1_1_1_1_0_1_1/symbolic | 3 | **slow / incomplete** | — | TO 3600 | — | 1022 | — | — | TO |
| 316 | finiteB_7_471_13_0__7_1_1_0_1_0_1_1_1/symbolic | 4 | **slow / incomplete** | — | TO 3600 | — | 3738 | — | — | TO |
| 317 | finiteBx12_7_471_13_0__7_1_1_0_1_0_1_1_1/symbolic | 4 | **slow / incomplete** | — | TO 3600 | — | 2782 | — | — | TO |
| 318 | finiteC_6_215_11_0__6_1_1_0_1_0_1_1_0/symbolic | 4 | **slow / incomplete** | — | TO 3600 | — | 600 | — | — | TO |
| 319 | finiteC_6_231_7_0__2_1_1_0_0_1_1_1_0/symbolic | 2 | **slow / incomplete** | — | TO 3600 | — | 1877 | — | — | TO |
| 320 | finiteC_6_470_10_0__0_5_1_0_1_0_1_1_1/symbolic | 6 | **slow / incomplete** | — | TO 3600 | — | 1173 | — | — | TO |
| 321 | finiteC_6_476_10_0__0_0_5_1_1_0_1_1_1/symbolic | 4 | **slow / incomplete** | — | TO 3600 | — | 2126 | — | — | TO |
| 322 | finiteC_7_247_10_0__4_1_1_0_1_1_1_1_0/symbolic | 3 | **slow / incomplete** | — | TO 3600 | — | 1031 | — | — | TO |
| 323 | finiteC_7_478_10_0__0_4_1_1_1_0_1_1_1/symbolic | 5 | **slow / incomplete** | — | TO 3600 | — | 728 | — | — | TO |
| 324 | finiteCx12_6_215_11_0__6_1_1_0_1_0_1_1_0/symbolic | 4 | **slow / incomplete** | — | TO 3600 | — | 1841 | — | — | TO |
| 325 | finiteCx12_6_231_7_0__2_1_1_0_0_1_1_1_0/symbolic | 2 | **slow / incomplete** | — | TO 3600 | — | 2347 | — | — | TO |
| 326 | finiteCx12_6_476_10_0__0_0_5_1_1_0_1_1_1/symbolic | 4 | **slow / incomplete** | — | TO 3600 | — | 2416 | — | — | TO |
| 327 | finiteCx12_7_247_10_0__4_1_1_0_1_1_1_1_0/symbolic | 3 | **slow / incomplete** | — | TO 3600 | — | 803 | — | — | TO |
| 328 | finiteCx12_7_478_10_0__0_4_1_1_1_0_1_1_1/symbolic | 5 | **slow / incomplete** | — | TO 3600 | — | 1143 | — | — | TO |
| 329 | finiteD_7_251_12_0__6_1_0_1_1_1_1_1_0/symbolic | 6 | **slow / incomplete** | — | TO 3600 | — | 824 | — | — | TO |
| 330 | finiteD_7_375_13_0__7_1_1_0_1_1_1_0_1/symbolic | 4 | **slow / incomplete** | — | TO 3600 | — | 220 | — | — | TO |
| 331 | test_gluza_ex33/solve/3 | 1 | **slow / incomplete** | — | TO 3600 | — | 179 | — | — | TO |

# Measurement notes and implementation

## Scope and comparability

The table contains all 331 exact polynomial/domain/order equivalence groups, including fixed and symbolic ZZ examples, core examples, and disabled upstream fixtures. “IDs” counts the retained original aliases; the accompanying `data.json` records every alias, input identity, source report and executable hash.

A row is complete only when the full original problem has an independently verified result. Completing its positive and negative branches separately does not automatically certify the full split problem. The current audit has 303 verified groups and 28 unresolved groups. The unresolved set is marked slow and assigned full one-hour campaigns. The first complete campaign exhausted all 28 budgets. Missing times and cells are not zeros, and timeouts do not produce finite speedup ratios.

The completed one-hour campaign (10 October 2026) timed out on all 27 symbolic ZZ originals and positive Gluza33. None exported a complete result or verification. These attempts add no completion credit; the frozen queue snapshot and all 28 ledger hashes are in the JSON. Targeted continuations retain completed algebra separately.

Measurements prefer verified three-repeat controls, otherwise the fastest accepted original-problem run. Builds are labelled. Cache tags mean cold (C), shared and possibly warm (S), or unrecorded (?). Solve times include in-run search, preprocessing and export. Earlier order discovery is separate; Gluza30 positive used a disclosed 300.026-second search. Verification has its own column.

Baseline ratios use the frozen `9025e71` release and original native GCAD, with identical inputs, orders, merge settings, CPU assignment and paired limits. For two sign branches, displayed times sum their separate three-repeat median phases; ratios compare those matched sums. They do not compare the optimized solve column with an unrelated native time. All 483 native-completing branches matched raw cell counts and independently verified; 96 native timeouts remain explicit. Geometric mean ratios are 0.15 for matched phases and 0.062 for whole processes.

New reference comparisons use at most max(300 s, 3 × completed symGCAD solve-wall median) per native repetition, subject to shorter explicit and campaign limits. Qualification runs use a 300-second ceiling before Rust timings are known. Native timeouts have no exact ratio; the historical baseline above retains its recorded limits.

## Machine and resource policy

AMD EPYC 9754 128-Core Processor, Linux 6.18.45, x86-64. The task budget is 16 CPU workers and 64 GiB aggregate memory; each measured solver process uses one CPU. Historical per-worker memory ceilings range from 4 to 32 GiB and are retained in the source ledgers. RAM is measured resident memory, not the configured ceiling. Three historical supplemental verification runs retain elapsed time but no RSS; their RAM entries carry † and report the available solve RSS only. Concurrent campaigns may share the host, so single-run differences are not presented as precise speedups. The ongoing slow campaign uses bounded process supervision, durable attempt records and independent proof verification.

**Supplemental sanity check.** Fixed A446 passed two fresh rational points in each of its eight exported cells (seed 20261010): 16 exact original-F evaluations and 112 strict-constraint checks, with zero failures or unchecked cells. Boundaries were recomputed at each fresh prefix. This separate check is outside benchmark timings and supplies no coverage-proof credit; full points and counterexamples from a deliberately wrong sign label are retained in `reference/SUPPLEMENTAL_SANITY.md`.

## What changed

**Exact algebra and reuse.** Rust uses Symbolica for polynomial arithmetic, factorization, resultants and rational root work. Canonical factors, deterministic algorithms and bounded caches avoid repeated work. Certified sum-variable preprocessing and primitive parameter-linear factors address measured algebra bottlenecks.

**Smaller valid decompositions.** Positive-domain certificates, homogeneous charts, conditional quadratic monotone reduction and strict linear/bilinear identities can avoid irrelevant projection factors or restore whole fibers from a smaller verified base. Fixed A381/Ax381 and the negative Gluza cases now have independently verified compact decompositions. Every specialized rule carries exact replayable evidence; ordinary projection remains a fallback.

**Bounded local and ordering work.** Local projection supports positive-ray certificates, finite witness stocks, strict Bézout pair separation and explicit resource stops. MCTS preserves variable groups and fixed-order obligations, uses bounded trial tiers and resumable journals, and admits winners only after fresh completion. Optional local progress scoring changes promotion priority, not mathematical proof acceptance.

## Reproduce and inspect

The Typst source and frozen `data.json` render without private literature or original proof files. Regeneration from retained evidence uses the Rust example `benchmark_report` its source records the selection rules and rejects missing identities or changed audited ledgers. Run `nix develop -c typst compile benchmark/benchmark.typ benchmark/benchmark.pdf`. Full provenance remains in the JSON snapshot and the repository’s benchmark/reference notes.